

The Road to Ubiquity: Unpacking Barriers to Mass Adoption of Heads-Up Computing

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ABSTRACT

We examine the trajectory and potential of heads-up computing technologies within the current computing landscape. Despite significant advancements and investments, this technology has yet to achieve widespread market penetration comparable to that of smartphones. Will these technologies remain niche products or do they have the potential to displace traditional mobile devices? To become fundamental in everyday life, we posit that both technological evolution and a deeper understanding of their utility and social integration are crucial.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous computing**; **Mobile computing**.

KEYWORDS

wearables, head-up displays

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1 INTRODUCTION

Head-Up Displays have been present in academia for well over half a century [17]. We have also seen several rounds of products come and go in the market and within the past decade there has been a lot of activity such as Google Glass, the Meta Quest, and most recently the Apple Vision Pro. However we have yet to see explosive adoption by the mass market as we did when Apple released the iPhone.

Prior to the iPhone, smartphones existed, but in a very different form. Pre-iPhone, smartphones were dominated by the RIM Blackberry, and both Motorola and Nokia had countless offerings. However once the iPhone was released and Android phones came rapidly afterwards, the landscape of smartphones changed radically with those previous incumbents all but dying out a few years later.

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This is a classic example of Christensen's theory of disruption in action [4].

Is heads-up computing going to remain a much smaller vertical within the larger landscape of computing as it is today? Or is there an opportunity for these technologies to replace the current dominance of mobile phones? The technical capabilities of smartphones such as extremely fast and low power processors, associated radios, cameras and battery improvements have been welcome improvements well aligned with heads-up computing platforms. Unfortunately the optical systems needed for head-up displays (HUDs) have not had the same level of investments and advancements brought about by the mobile revolution. On the interaction side, novel sensing and low power computer vision and speech recognition have opened the door for new ways to interact with these devices, but this is an area where there is still a tremendous amount of exploration with no clear contender as there was with the iPhone and the capacitive touch screen, or the keyboard and mouse and associated direct manipulation paradigm from the personal computing era [7]. While both the hardware platforms, and associated interaction paradigms are active areas of research, there are still significant open questions around usefulness that need to be answered: what value is the computing platform bringing in everyday life and how might these devices complement humanity as deeply social beings? In this paper, we outline these key areas for needed future research.

2 FINDING USEFULNESS

What are the capabilities we are gaining with heads-up computing especially in light of the dominance of mobile phones? This still seems to be a very much open question and the notion of what constitutes "better" is unclear. Today's mobile phones are significantly worse in many ways relative to desktop computing counterparts with slower text entry, less precise pointing, smaller displays, etc. Instead it was their ability to always be with us both as computational and communication (Internet) platforms and usable in time frames that are good enough [1]. Compared to the previous generation of smartphones, the iPhone started competing on different metrics that were previously not addressed which led to its ultimate success (a new market disruption in Christensen's theory [3]). It seems unlikely that heads-up computing platforms are going to out-compete smartphones on a trajectory of being better than them. Instead, they need to find new uses and previously unmet needs within the general population to really take off.

This idea was present when Google released Glass with the Explorer Edition with the deliberate idea of recruiting early adopters to explore what Glass might be useful for. For a variety of reasons, that exploration did not play out and Glass' core capabilities became notifications and a wearable camera. Notifications are present on

smartphones and smartwatches, and these features seem unlikely to be a compelling driver of adoption. The camera capability within Glass gained unwanted notoriety which we will touch on below, and itself competed with the camera in smartphones, or wearable action cameras like the GoPro. If we go back farther in time, the early researchers who lived with these systems did so in an academic setting. The technology became useful as a laboratory notebook [9] in the highly mobile setting of universities and conferences where they served as a memory prosthesis of sorts. Applications like the Remembrance agent [15] used by some of these early wearable users were in the vein of tools to augment scientific thought that came before [2, 5, 8]. While compelling to academics, this use case is unlikely to be valued by the public at large.

When Google revamped the Glass program they shifted towards industrial use cases which is not surprising given a lot of the early wearable computing research was directly in this domain such as aircraft maintenance [16] and food processing plants [13] amongst many others. While potentially very valuable in these settings, this is a quite different from mass adoption.

Today's mixed reality headsets seem to have found a foothold in immersive gaming and media experiences in particular. However we have yet to see much adoption outside of these domains. So while these devices might offer the next evolution for gaming consoles such as the Microsoft Xbox or Sony PlayStation, they do not yet seem poised to overtake the mass market of mobile phones. In other words, the modern mixed reality head-up computing platforms might also find success in situated usage, but do not seem positioned to deeply enter our lives in the way smartphones have.

For example, if you were to leave home in a hurry, you would grab your phone, your wallet and your keys. Or a more extreme example, if you were to go out to the club for the night, that might be paired down to just your phone (using digital payments and a digital ID). What would it take for our future heads-up computing platform to be selected on the way out for the night instead? Or if we are out for dinner and drinks with friends, someone might pull out their phone for a few moments - would they don a heads-up computing platform in the same social setting?

3 CONSIDERING THE SOCIAL

The social, or anti-social, nature of heads-up computing devices seems like the other key challenge to address before these technologies see large scale adoption. A lot of research understandably has gone into understanding how the device augments its wearer towards the argument in the section above. However we are social beings living in a social setting and the current view of these technologies is extremely egocentric. An augmented reality system might augment the senses and capabilities of the wearer, but what of others around that individual? Do they have any idea of the augmentations experienced? Right now the answer is largely no.

This is one of the challenges Google Glass faced with the camera. While the camera was only used momentarily in general, any onlooker only knew the camera was physically present all the time when someone was wearing Glass. While early camcorders had a blinking red LED (in some countries mandated by law) it is not clear that is the right solution for these types of devices. As such, socially signaling that the device is actively being used or not is one key

consideration for these devices that has been largely unexplored and was one of our explicit design goals in creating Loupe [10]. The Apple Vision Pro with the virtual display of the user's eyes does provide some signaling to others around, but the fidelity and awkwardness of the display seems to limit utility.

Beyond signaling if the device is being used, what is it being used for? These are new technologies and non-users have poor mental models for usage. People assumed Google Glass users were always taking photos. In contrast, if someone has a phone out at dinner, the others at the table might not know exactly what the person is doing, but from their own usage they have a sense of what the options are. This might just be a communication and understanding challenge during the formative adoption of a given technology. Or given the above issues, it might be more fundamental. An example from the past of a success is the introduction of the Sony Walkman and its headphones. During the public debut, Sony hired roller skaters to wear this very new kind device and skate around a park in Tokyo. They deliberately invited the press to see the devices in action [12]. Sony very carefully crafted the presentation of the new technology. And while not a tool used in academia, the careful planning of a public relations (PR) campaign and associated marking can be used to guide perceptions.

A related challenge is these devices live in our intimate space [6], often in direct contact with our body. While one might turn a laptop to show a colleague their screen, or pass their mobile phone to a friend to view, doffing one's wearable for another to don would be awkward at best; and, instead of creating a shared experience, it just shifts the computational capabilities from one individual to another.

There has been some research on the social acceptability of wearables (e.g. [14]). However, this is likely a complex space that needs to be teased apart more. Being disruptive is likely a bad thing, although it did not prevent the adoption of Bluetooth headsets. While hearing only one side of a conversation of someone using one is not all that socially graceful, it does offer transparency. Once a passerby determines they are not being spoken to, they do have good insight into the nature of the behavior. How do we make our interactions with heads-up computing devices overt yet non-disruptive? This is a property that current computing technologies offer in general. Onlookers can see when someone is using a laptop or phone and can guess what they might be doing with the devices.

Finally, how might we leverage heads-up computing technologies to not only augment the user, at minimal cost in a social setting but instead to enhance face-to-face social interactions? How can the technology support the wearer in a conversation to make the conversation better? For example, early work on dual purpose speech was in this vein [11]. Or how might it support others both with and without their own instances of the technology and enhance in person communication and associated social interactions? There is much research to be done in this area. More broadly, how can we shift from an egocentric view of support to one that is much more cognizant of the physically instantiated social world we live in?

4 CONCLUSION

While heads-up computing technologies have offered new paradigms in computing, their mass adoption remains uncertain. Unlike the

seismic shifts seen with the advent of smartphones, they have not yet found their “iPhone moment,” where they offer an indispensable value that redefines user interactions or fulfills unmet needs propelling them to mass adoption. The challenges lie not only in technological advancements, but in defining and demonstrating their utility in everyday contexts. To transition from vertical solutions for specific use cases into integral parts of our daily lives, further exploration into socially grounded uses and interactions is essential. This will require not only continuous technological evolution but also a deeper understanding of how these devices can seamlessly integrate into and enhance our social fabric, moving beyond individual augmentation to enrich in person collective experiences.

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